2025 Incident Review Summary



“If you’re not learning, you’re wasting your time.”

Pharrell Williams

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Introduction

The information in this report comes from wildland fire incidents—from various entities—submitted to and gathered by the Wildland Fire Lessons Learned Center (LLC) in 2025. The primary source of this data is incident reports (Rapid Lesson Sharing, Facilitated Learning Analyses, Serious Accident Investigations, etc.). Most of these reports have been posted to the LLC’s Incident Review Database. SAFENETs and other data sources have been included when no full report was produced or could not be located.

In 2025 we collected information on over 130 operational incidents.

We have examined these incident reports and learning documents, compiled a few numbers, and extracted specific lessons. It is our aim to present this information in a way that empowers us all to improve future operations. We hope you turn these lessons into learning.

Exercises

This icon identifies specific discussion prompts related to the material.



Fatalities

In 2025, the Wildland Fire Lessons Learned Center recorded **12** wildland fire-related fatalities, 1 more than 2024.

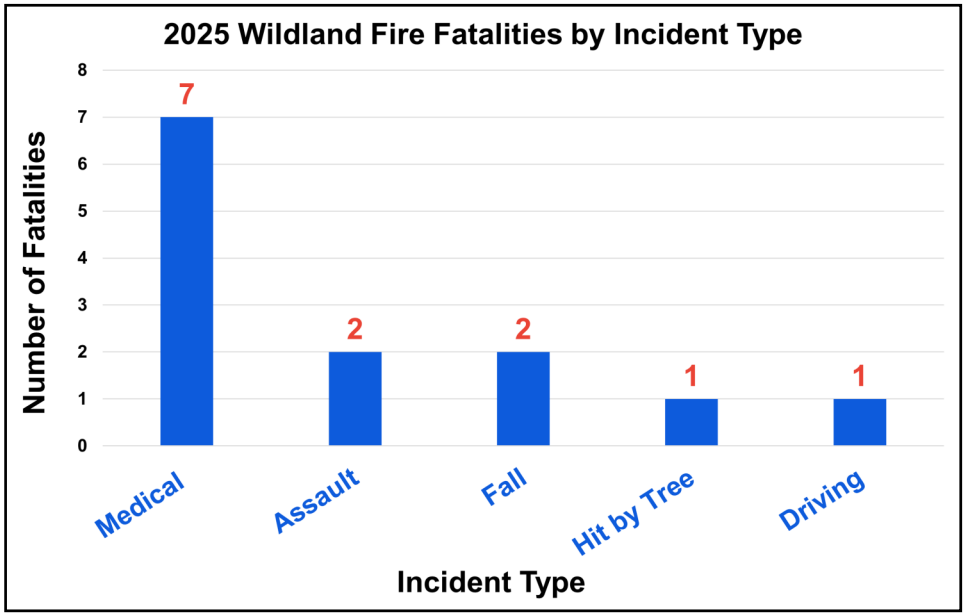
Exercise:
Use the following prompts for discussion:

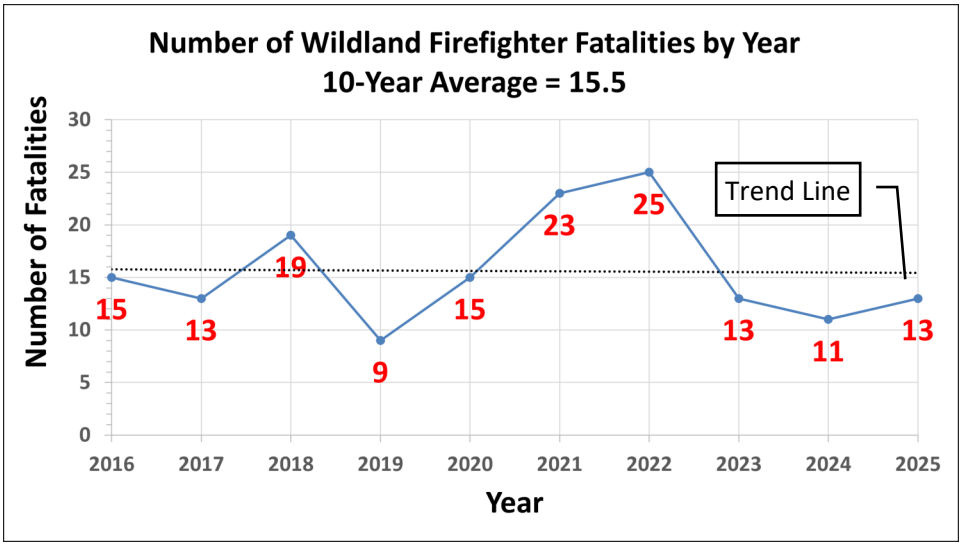
- What in your training prepares you for an assault type of situation?
- Based on your role, in what situations are you most exposed to a deadly fall?



Fatalities: 13

- 7 Medical
- 2 Falls
- 1 Hit by Tree
- 2 Assault
- 1 Driving





We have recorded a total of 155 wildland fire-related fatalities over the past ten years.

Chainsaw Cut Incidents

In 2025, 55 percent of the chainsaw cut incidents involved kickback. Kickback is a well-known phenomenon and is often discussed in safety briefings for chainsaw operations. Is there a way to improve the effectiveness of these briefings?

Exercise:
Use the following prompts for discussion:

- What have you been taught about chainsaw kickback?
- What is the best way to learn about the dangers of kickback?





11 Chainsaw Cut Incidents

6 Involved Kickback

Who got cut?
- 8 sawyers
- 3 swampers

"Keep it simple; minimize complexities." Oregon Project Work Chainsaw Cut 2025

Body part cut:
- 7 to the hand
- 2 to the leg
- 1 to the foot
- 1 to the shoulder

"Be aware of body placement and how that correlates with potential kickback." Kaibab National Forest Chainsaw Kickback Laceration

What was the task?
- 4 tree felling
- 4 brushing
- 3 limbing/bucking

"It would have been better to work with a swamper and lock the chain brake before clearing the debris from the bar." South Rim Fire Chainsaw Cut

Chaps Work

In addition to the 11 chainsaw cut injuries, we also recorded two instances of chap strikes with no injuries.

"These chaps did what they were made to do and stopped the saw from cutting into my leg."
Texas Storm Damage Chainsaw Chaps Cut



Hit by Tree Incidents

Numerous firefighters get hit by trees every year. On multiple occasions—including during 2025—the primary supervisor in a group is the person struck. When this occurs, it requires others to take charge in an emergency situation.

Exercise:
Use the following prompts for discussion:

- What can you do to prepare for taking charge during an emergency?
- What are some potential complications of describing your location on a fire?





26 Hit by Tree Incidents

- **12** During Tree Felling Ops
- **14** During Non-Tree Felling Ops

Of the **12** tree felling operations resulting in tree strikes:
- In **10** instances the sawyer was hit.
- In **2** instances someone other than the sawyer was hit.

Of the **14** non-tree felling operations resulting in tree strikes:
- **12** occurred during fire suppression work.
- **2** occurred during prescribed fire work.

Sleeping Area Tree Strike

The Division Supervisor and field Safety Officer noticed a snag adjacent to the camping area. A discussion took place regarding this hazard; however, they did not have the means to eliminate it. At 4 a.m. the snag fell and struck a sleeping firefighter.

- Take the time to clear all hazards from sleeping areas, regardless of the effort required to complete the task.

Aggie Creek Fire Spike Camp Hit by Tree 2025

There was one Hit by Tree fatality in 2025.
The fatality occurred on a prescribed fire in Idaho.
Tinker Bugs Prescribed Fire Hit by Tree Fatality

Driving-Related Incidents

We recorded 27 separate driving-related incidents in 2025.

Driving decisions can often be influenced by external pressures. Homes at risk, critical burn operations, or initial attack response time pressure all impact our risk assessments and can change our focus while driving.

Exercise:
Use the following prompts for discussion:

- How do your driving decisions differ when driving to an initial attack fire, a project fire, or back to your home unit?
- While supporting operations on a fire (such as pump and roll, pre-treating, fuel shuttle), is your primary focus still on driving?



Incident Types

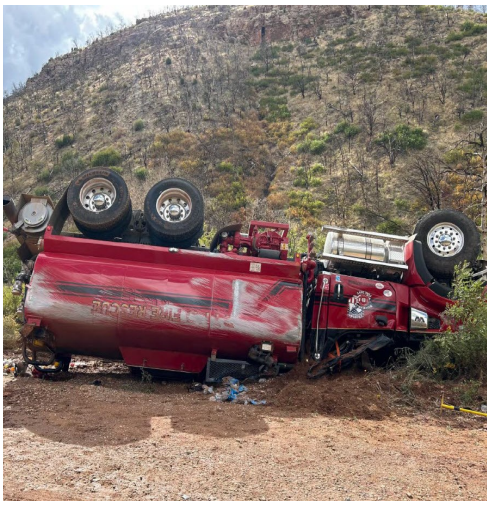
- 15 Rollovers
- 7 Crashes
- 2 Backing
- 2 Burn Damage
- 1 Broken Wheel Studs

Vehicles involved: engines, crew carriers, UTVs, pickups, a water tender, a septage truck, and a heavy equipment transport.



There was one driving-related fatality in 2025 (water tender rollover). The fatality occurred in South Dakota on July 26.

War Creek Fire Water Tender Accident Fatality 2025



Rollovers

In 2025, **15** of the driving-related incidents involved rollovers**: 7** UTV and **8** vehicle. Lessons centered on scanning the area in front of the moving vehicle for hazards, trail/road features, and changing conditions. Consider additional weight factors caused by the modifications to your UTV when operating off-road.



Lug Nuts

In 2025, one incident involved lug nuts not being properly torqued resulting in the loss of two tires.

Consider marking lug nuts and studs with paint, or using plastic lug nut position indicators to visually detect any loosening.

Entrapments

In 2025, all entrapments involved firefighters in vehicles.

Exercise:
Use the following prompts for discussion:

- Do you have a greater sense of safety or control being in a vehicle vs. on foot?
- How does a vehicle make you more susceptible to getting in a "bad spot"? How can you fight this tendency?









"Deceptively light flashy fuels burn hot and move quickly. Frontal assault on a fire is a 'Watch Out' and is a tactic to be utilized in limited situations. Attack from the black or keeping one foot in the black is the safest tactic."

West Beach Prescribed Fire Entrapment

In 2025, the LLC recorded 5 entrapments, significantly fewer than the 14 recorded in 2024. The entrapment situations reported were more evenly split between Initial Attack, Extended Attack, and Prescribed Fire than in 2024, when Initial Attack was the most common incident stage:

- 2 during Extended Attack
  Both incidents involved dozers in New Mexico
- 2 during Initial Attack
  Both incidents involved engines
- 1 during Prescribed Fire
  A UTV suppressing a spot fire

Trout Fire Dozer Entrapment

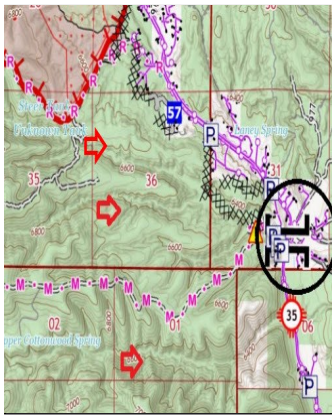
While cutting off a running slopover in grass and juniper, flames impacted the open-cab dozer. The operator ran to safety, sustaining burns on his hands and head in his escape.

Conner Fire Entrapment

On an initial attack incident, the assistant engine captain was charging a hose when flames impacted him and the engine.

Lessons from the Learners

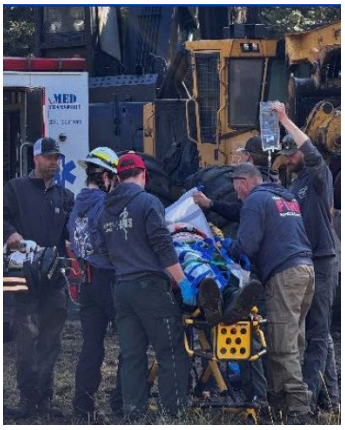
This section includes lessons from those who were directly involved in the incidents.



[Trout Fire Dozer Entrapment]

During an assignment we develop shorthand and slang to refer to landmarks. These names might be related to where something happened three days ago, a local name, or a name used on Google Maps—but not printed on incident maps. Remember that people may need to find a location using only an Ops map. Drop points and other landmarks are placed on the map to create a shared geographic vocabulary.

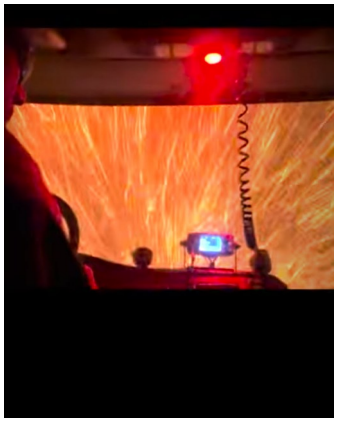
Lesson: In an emergency, describe locations using mapped landmarks. When communicating locations, especially to resources who haven't been in the area every day, use names found on incident maps and IAPs.



[Pomas Fire Fall Injury Extraction]

A Tender Operator called on a tactical channel: "I think I broke my ribs. I need help." A Line EMT heard the traffic and asked for the location. The Tender Operator said he was at the Schaefer Creek fill site. A Safety Officer also asked for more location details, such as a drop point. Fortunately, the EMT recognized the sign for Schaefer Creek Campground and confirmed the location. The incident map did not identify the fill site with a specific name or designated drop point. The important landmark was inadvertently obscured with pump symbols or field site icons, which hindered navigation and coordination.

Lesson: Use distinct, clearly labeled landmarks or features when dropping pins or icons on digital maps. Do not obscure existing labels.



[Deer Creek Fire Pyro-Vortex Incident]

The engine that decided to stay in place at the structure made that decision based upon scouting the area the day before and determined it had the appropriate level of defensible space, even with the conditions they had previously seen. Though the accompanying tornado caused the brush to burn and structure to catch fire, the engine did remarkably well with the extreme fire activity and impact of an EF2 tornado.

Lesson: Have a plan and know where your people are.
This event reinforced the need to continuously assess the fire activity, verify your resources are in a good location, and ensure everyone has a backup plan.

Lessons from the Learners

This section includes lessons from those who were directly involved in the incidents.



Olympic NF Vehicle Accident

A trailer with a load of logs detached from its tow vehicle and rolled perpendicularly across both lanes of the highway—directly into the path of the USFS vehicle. The log truck's loaded trailer and USFS vehicle struck each other in the ditch on the highway's north side. The airbags deployed and logs from the trailer entered through the windshield. Both occupants walked away with only minor injuries.

Lesson: Safety mitigations make a big difference.
This was a newer vehicle with advanced airbags. The airbags, seatbelt use, and defensive driving were critical elements in this outcome.



Billy Fire UTV Accident

The rear tires sank off the roadbed's steep edge. Unable to gain traction, the UTV tilted, slid, and impacted something hard. The UTV and crew somersaulted backward down into the drainage. The crew, all wearing hardhats, hit the UTV ceiling as unrestrained bodies and gear tumbled in different directions.

Lesson: Wear a DOT helmet and use the seatbelt.
Firefighters often do not wear the U.S. Department of Transportation (DOT) helmets and seatbelts in UTVs. Roads and terrain that appear of low consequence can result in high-severity injuries when UTV occupants do not wear seatbelts and leave the Rollover Protection System (ROPS) without wearing a DOT helmet.



Ninetyeight Fire Boating Accident

Five firefighters working on the Ninetyeight Fire were injured in a boating incident in which two boats collided and the firefighters were ejected into the water. This collision between two watercraft has highlighted the urgent need for improved traffic coordination, communication, and visibility in boating operations.

Lesson: When using boats of any kind, evaluate risk and plan for emergencies specific to boat operations.
Boats are regularly used to transport firefighters. If you are participating in any boat operations, consciously think through what an emergency on the water will require.

20 Years of Assault-Related Incidents

Incidents involving an active shooter or threatening behavior have occurred in a variety of locations including along active fireline, at helibase, at work stations, and on patrol.

Exercise:

Use the following prompts for discussion:

- Based on your job, in what environment are you most likely to encounter an active shooter?
- What are some common fire-related scenarios that can lead to confrontation with the public?



In June of 2025, a horrific event unfolded during initial attack operations in Coeur d'Alene, Idaho. Three firefighters were shot. Two of them died.

We found 10 assault-related events involving wildland fire personal over the past 20 Years.
90% Involved Firearms

Nettleton Gulch Shooting Fatalities
Job Corps Center Shelter in Place
Bootleg Fire Shooting
Active Shooter at Helibase
Los Angeles County Shooting Fatality
Miriam Fire Shooting Incident
Tepee Springs Fire Confrontation
Assault at Remote Work Station
Engine Patrol Gunshots
Gooding Gun Club Fire Confrontation

"Incident Management Teams throughout the country have seen an uptick in hostility and frustration from the public during their assignments. In several of these incidents, IMTs have felt the need to be proactive and develop active shooter plans."— 2021 RLS Active Shooter Awareness for Incident Management Teams



A man pulled a handgun out and pointed it at the Helibase Manager. Derek, the Squad Leader, took shelter and called 911. One crew member calmly walked from pad to pad and told people there was a man with a gun.
Active Shooter at Helibase

The steps outlined below were developed by the Helitack Crew involved in this incident.

Threat Planning

1. Where are you operating? (airport, remote site, urban setting, etc.)
2. What steps can you take to limit access to your location?
3. Have you identified a rendezvous point?
4. What is your communication plan and how will authorities be notified in an emergency?
5. Is an alert word established and known?
6. What is your plan of action? Remain as calm as possible. Alert adjacent resources of potential threat. Evacuate, warn, assist, and direct. RUN, HIDE, FIGHT.